KAIZEN MATHS

Lesson starter

Quadratic Equations

Quadratic Formula

Name _____

S1

$x^2 + 7x - 3 = 2$

S2

$4x^2 - 2x - 5 = 12$

S3

$x^2 - 6x + 4 = 2$

S4

$2x^2 - 4x - 2 = 7$

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Exit ticket

Quadratic Equations

Quadratic Formula

Name _____

E1

$x^2 + 2x + 6 = 3$

E2

$4x^2 - 4x + 6 = 3$

E3

$x^2 - 2x - 9 = 7$

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Exit ticket

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Quadratic Formula

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$x^2 + 2x + 6 = 3$

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$4x^2 - 4x + 6 = 3$

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Quadratic Formula

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$x^2 + 2x + 6 = 3$

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Quadratic Formula

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$x^2 + 2x + 6 = 3$

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$x^2 - 2x - 9 = 7$

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Teacher answers & working

Quadratic Equations · Quadratic Formula

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S1

$x^2 + 7x - 3 = 2$

Answer:

$x_1 \approx 0.65, x_2 \approx -7.65$

Working:

Step 1: Rearrange to $ax^2+bx+c=0$ and use the quadratic formula.

Step 2: Rearrange to standard form:

$x^2 + 7x - 5 = 0$

Step 3: Use the quadratic formula: $x = \frac{-b \pm \sqrt{(b^2-4ac)}}{2a}$.

Here a = 1, b = 7, c = -5.

Step 4: Discriminant: $D = b^2 - 4ac = 7^2 - 4 \times 1 \times -5 = 69$.

Step 5: Substitute into the quadratic formula:

$x_1 = \frac{-7+\sqrt{(69)}}{2\times1} \approx 0.65$

$x_2 = \frac{-7-\sqrt{(69)}}{2\times1} \approx -7.65$

Step 6: Final answer: $x_1 \approx 0.65, x_2 \approx -7.65$

S2

$4x^2 - 2x - 5 = 12$

Answer:

$x_1 \approx 2.33, x_2 \approx -1.83$

Working:

Step 1: Use the quadratic formula with a coefficient of x^2 greater than 1.

Step 2: Rearrange to standard form:

$4x^2 - 2x - 17 = 0$

Step 3: Use the quadratic formula: $x = \frac{-b \pm \sqrt{(b^2-4ac)}}{2a}$.

Here a = 4, b = -2, c = -17.

Step 4: Discriminant: $D = b^2 - 4ac = (-2)^2 - 4 \times 4 \times -17 = 276$.

Step 5: Substitute into the quadratic formula:

$x_1 = \frac{2+\sqrt{(276)}}{2\times4} \approx 2.33$

$x_2 = \frac{2-\sqrt{(276)}}{2\times4} \approx -1.83$

Step 6: Final answer: $x_1 \approx 2.33, x_2 \approx -1.83$

Teacher answers & working

Quadratic Equations · Quadratic Formula

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S3

$x^2 - 6x + 4 = 2$

Answer:

$x_1 \approx 5.65, x_2 \approx 0.35$

Working:

Step 1: Rearrange to $ax^2+bx+c=0$ and use the quadratic formula.

Step 2: Rearrange to standard form:

$x^2 - 6x + 2 = 0$

Step 3: Use the quadratic formula: $x = \frac{-b \pm \sqrt{(b^2-4ac)}}{2a}$.

Here a = 1, b = -6, c = 2.

Step 4: Discriminant: $D = b^2 - 4ac = (-6)^2 - 4 \times 1 \times 2 = 28$.

Step 5: Substitute into the quadratic formula:

$x_1 = \frac{6+\sqrt{(28)}}{2\times1} \approx 5.65$

$x_2 = \frac{6-\sqrt{(28)}}{2\times1} \approx 0.35$

Step 6: Final answer: $x_1 \approx 5.65, x_2 \approx 0.35$

S4

$2x^2 - 4x - 2 = 7$

Answer:

$x_1 \approx 3.35, x_2 \approx -1.35$

Working:

Step 1: Use the quadratic formula with a coefficient of x^2 greater than 1.

Step 2: Rearrange to standard form:

$2x^2 - 4x - 9 = 0$

Step 3: Use the quadratic formula: $x = \frac{-b \pm \sqrt{(b^2-4ac)}}{2a}$.

Here a = 2, b = -4, c = -9.

Step 4: Discriminant: $D = b^2 - 4ac = (-4)^2 - 4 \times 2 \times -9 = 88$.

Step 5: Substitute into the quadratic formula:

$x_1 = \frac{4+\sqrt{(88)}}{2\times2} \approx 3.35$

$x_2 = \frac{4-\sqrt{(88)}}{2\times2} \approx -1.35$

Step 6: Final answer: $x_1 \approx 3.35, x_2 \approx -1.35$

Teacher answers & working

Quadratic Equations · Quadratic Formula

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E1

$x^2 + 2x + 6 = 3$

Answer:

No real solutions

Working:

Step 1: Rearrange to $ax^2+bx+c=0$ and use the quadratic formula.

Step 2: Rearrange to standard form:

$x^2 + 2x + 3 = 0$

Step 3: Use the quadratic formula: $x = \frac{-b \pm \sqrt{(b^2-4ac)}}{2a}$.

Here a = 1, b = 2, c = 3.

Step 4: Discriminant: $D = b^2 - 4ac = 2^2 - 4 \times 1 \times 3 = -8$.

Step 5: The discriminant is less than 0, so there are no real solutions.

Step 6: Final answer: No real solutions

E2

$4x^2 - 4x + 6 = 3$

Answer:

No real solutions

Working:

Step 1: Use the quadratic formula with a coefficient of x^2 greater than 1.

Step 2: Rearrange to standard form:

$4x^2 - 4x + 3 = 0$

Step 3: Use the quadratic formula: $x = \frac{-b \pm \sqrt{(b^2-4ac)}}{2a}$.

Here a = 4, b = -4, c = 3.

Step 4: Discriminant: $D = b^2 - 4ac = (-4)^2 - 4 \times 4 \times 3 = -32$.

Step 5: The discriminant is less than 0, so there are no real solutions.

Step 6: Final answer: No real solutions

Teacher answers & working

Quadratic Equations · Quadratic Formula

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E3

$x^2 - 2x - 9 = 7$

Answer:

$x_1 \approx 5.12, x_2 \approx -3.12$

Working:

Step 1: Rearrange to $ax^2+bx+c=0$ and use the quadratic formula.

Step 2: Rearrange to standard form:

$x^2 - 2x - 16 = 0$

Step 3: Use the quadratic formula: $x = \frac{-b \pm \sqrt{(b^2-4ac)}}{2a}$.

Here a = 1, b = -2, c = -16.

Step 4: Discriminant: $D = b^2 - 4ac = (-2)^2 - 4 \times 1 \times -16 = 68$.

Step 5: Substitute into the quadratic formula:

$x_1 = \frac{2+\sqrt{(68)}}{2\times1} \approx 5.12$

$x_2 = \frac{2-\sqrt{(68)}}{2\times1} \approx -3.12$

Step 6: Final answer: $x_1 \approx 5.12, x_2 \approx -3.12$